
Climate Resilience for Vulnerable Districts

Designing Community Flood Early Warning Systems and Nature-Based Flood Protection that Deliver Measurable Risk Reduction and Can Be Scaled Across Multiple Districts

*A PhD-level research brief and implementation framework
for an India-focused, practitioner-scholar inquiry*

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*Anchored in IPCC AR6, UNDRR Sendai Framework, WMO Multi-Hazard EWS guidelines, and the
UN Early Warnings for All initiative*

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Prepared as a structured, multi-stage analysis. Reasoning is made explicit throughout. Where evidence is contested or uncertain, that uncertainty is named rather than hidden.

A. Executive Overview

Floods are the single most frequent and most costly hydrometeorological hazard in India and globally. The IPCC Sixth Assessment Report (AR6, WGII, 2022) is unambiguous that climate change is intensifying extreme precipitation in monsoon Asia, that exposure is rising due to settlement and asset growth in floodplains, and that vulnerability remains stubbornly high for the poorest social groups. The UNDRR Sendai Framework Midterm Review (2023) and the UN Secretary-General's "Early Warnings for All" (EW4All) initiative (launched 2022, with an implementation plan to 2027) together establish a global policy expectation: every person on Earth should be protected by a multi-hazard early warning system, with nature-based solutions (NbS) embedded in disaster risk reduction (DRR) and adaptation portfolios.

Yet, in India's most vulnerable districts — across the Brahmaputra and Barak basins of Assam, the Kosi-Ganga floodplain of north Bihar, the eastern Uttar Pradesh terai, the Mahanadi delta in Odisha, coastal West Bengal, and several rapidly urbanising basins (Chennai, Bengaluru, Mumbai, Hyderabad) — the *operational* gap between national flood forecasting (run by the Central Water Commission and IMD) and *actionable, last-mile, community-level early warning* remains wide. Simultaneously, NbS for flood risk reduction (catchment treatment, wetland and floodplain restoration, riparian buffers, mangrove and bioshield reforestation, soil and moisture conservation) are deployed in fragmented, project-by-project ways with weak monitoring of hydrological co-benefits. Funders increasingly demand *measurable* risk reduction; current programs largely report inputs and outputs (volunteers trained, structures built) rather than risk-reducing outcomes (avoided losses, reduced exposure, lead time delivered, who was warned and who acted).

This brief reformulates the original problem as a tractable PhD inquiry plus a practice-oriented implementation question, then synthesises global and Indian evidence on community flood EWS and NbS. It builds a conceptual framework anchored in the IPCC hazard–exposure–vulnerability risk framing, the Sendai Framework's four priorities, and the WMO's four EWS components (risk knowledge, monitoring & forecasting, dissemination & communication, response capability). It then operationalises "measurable risk reduction" through a multi-tier indicator set that is acceptable both to peer-reviewed publication and to climate funders such as the Green Climate Fund (GCF), Adaptation Fund (AF), the World Bank (NCRMP, NHP, R-DRIP), Indian central programs (NDMA, NDRF, NAFCC, MGNREGS, NMSA, Jal Shakti Abhiyan, Namami Gange), state DMAs, and CSR/philanthropic vehicles.

The core argument is fourfold. **First**, neither EWS alone nor NbS alone is sufficient: EWS without NbS treats only the tail of the risk curve (warning before the event), while NbS without EWS leaves residual extreme-event risk unmanaged. An *integrated* EWS + NbS + institutional strengthening (IES) model — what this brief calls the **Integrated Last-Mile Resilience Model (ILRM)** — is theoretically and empirically more defensible. **Second**, the binding constraints in India are rarely scientific (forecasts have improved markedly with IMD's IFLOWS, NWM, GFS-based ensembles, and CWC's basin models); they are *institutional and last-mile* — the translation of a forecast into a household-level decision in a Mishing village in Majuli or a Musahar tola in Saharsa. **Third**, "scalability" has a precise meaning in this domain: a *replication architecture* (standardised protocols, open data, modular hardware, district disaster management plan integration, recurring public budget lines) rather than ad-hoc project replication. **Fourth**, funder attractiveness depends less on novelty and more on (a) a credible theory of change, (b) a measurable risk-reduction logframe with avoided-loss accounting, (c) co-benefits across SDGs 6, 11, 13, 15, and (d) demonstrable institutional pathways for sustained finance after the project ends.

For the PhD scholar, the most defensible design is a **comparative-plus-action-research hybrid in two contrasting Indian districts** — for example, one Brahmaputra-basin district (e.g., Dhemaji or Majuli, Assam: high recurrence, riverine, mature SDMA presence) and one Kosi-basin district (e.g., Supaul or Madhubani, Bihar: extreme recurrence, breach-driven, weaker last-mile). The minimum

viable research design uses open data (IMD gridded rainfall, CHIRPS, GPM-IMERG, ERA5, Sentinel-1 SAR for flood extent, SRTM/ALOS for DEM), open-source models (HEC-RAS, LISFLOOD, SWAT, HEC-HMS, Google Earth Engine for change detection), and low-cost community-EWS technologies (multi-channel SMS/IVR/WhatsApp cascades, sirens, community gauge readers, women-led volunteer networks). The four-year plan in Section D translates this into year-by-year deliverables compatible with part-time scholarship and practitioner work.

The brief closes by naming open research gaps suitable for a PhD contribution: (i) causal attribution of NbS hydrological effects at watershed scale under non-stationary climate, (ii) inclusion-weighted indicators of EWS reach (warning equity, not warning coverage), (iii) governance models for sustained district-level financing of integrated EWS+NbS, (iv) climate-change-adjusted cost–benefit frameworks for NbS that internalise ecosystem co-benefits, and (v) the "warning-to-action" behavioural gap, particularly for women, Dalit, and Adivasi households in flood-recurrent districts.

B. Detailed Sections

Section 1. Sharpening the Research Problem

1.1 From a broad theme to PhD-grade research questions

The original problem statement — designing community flood EWS and NbS that deliver measurable risk reduction and can be scaled — bundles a normative goal, a design challenge, and a measurement challenge. For a PhD inquiry, this must be decomposed into questions that are *answerable*, *bounded*, *contestable*, and *empirically tractable*.

Scientific / academic research questions (RQs). These ask what is true, measurable, or comparable.

- **RQ1 (Effectiveness and attribution).** *Under monsoonal flood regimes in vulnerable Indian districts, to what extent and through which mechanisms does an integrated community flood EWS + NbS intervention reduce flood risk (in terms of exposure, vulnerability, and avoided losses) compared to (a) EWS-only, (b) NbS-only, and (c) business-as-usual configurations?* This is a comparative-effectiveness question with explicit counterfactuals.
- **RQ2 (Last-mile equity and warning-to-action).** *Which design features of community EWS — channel mix, message framing, trusted intermediaries, response infrastructure — most strongly determine whether a warning is received, understood, trusted, and acted upon by the most vulnerable sub-groups (women, Dalit, Adivasi, elderly, persons with disabilities, landless tenants), and how does this last-mile efficacy moderate aggregate risk reduction?* This is a mechanism question, foregrounding equity.
- **RQ3 (Scalability architecture).** *What governance, financing, data, and protocol arrangements are necessary and sufficient to scale an integrated EWS + NbS model from one or two pilot districts to a state-wide or basin-wide program, and how do these arrangements perform under fiscal, climatic, and political stress tests?* This is an institutional-design question.

Practice-oriented implementation question (IQ).

- **IQ1.** *What is a minimally viable, funder-attractive package — a "blueprint" of hardware, software, community processes, NbS interventions, indicator system, and financing pathway — that a state government or a state–NGO–community consortium in India can adopt for a vulnerable district, and which can be replicated across at least 10 districts within 5 years using existing public schemes (NDMA, MGNREGS, Jal Shakti Abhiyan, NAFCC) blended with multilateral and CSR finance?*

The separation matters. RQ1–RQ3 are what the PhD *contributes to knowledge*. IQ1 is what the PhD *delivers to practice* and is the natural interface with funders. The dissertation can structure chapters such that each empirical chapter answers one RQ and the synthesis chapter answers IQ1 — a "thesis with impact" approach increasingly accepted in Indian universities (e.g., IIT-Roorkee, TERI SAS, IIIT-Delhi, IIT-Gandhinagar) and globally (UCL, Wageningen, IHE Delft).

1.2 Defining key terms — anchored in standard frameworks

Definitions must do double duty: precise enough for academic discourse and recognisable to funders.

Community flood early warning system (CFEWS). Building on the WMO and UNDRR consensus that an EWS comprises four interacting components — (i) disaster risk knowledge, (ii) detection, monitoring, analysis and forecasting of hazards, (iii) warning dissemination and communication, and (iv) preparedness and response capabilities — a *community* EWS is one in which the community is not merely the recipient of warnings but a co-producer of risk knowledge, a co-operator of monitoring (e.g., volunteer gauge readers, citizen rainfall observers), a co-designer of communication channels (language, framing, trusted messengers), and the primary agent of response. The Sendai Framework's

Priority 4 ("enhance disaster preparedness for effective response") and the Early Warnings for All initiative (UNDRR–WMO–ITU–IFRC, 2022) anchor this definition. Crucially, "community" here is *socially differentiated*: women, Dalits, Adivasis, tenants, migrants, and persons with disabilities have distinct information needs and response constraints, and a CFEWS that does not internalise this differentiation will under-perform on equity.

Nature-based flood protection / nature-based solutions (NbS). The IUCN Global Standard for NbS (2020) and the UNEA-5 resolution (2022) define NbS as actions to protect, sustainably manage, and restore natural or modified ecosystems that address societal challenges effectively and adaptively, providing human well-being and biodiversity benefits. For floods specifically, NbS includes upstream catchment treatment (afforestation, soil-and-moisture conservation, contour bunding, check dams of porous design), wetland and tank restoration, floodplain reconnection (room-for-the-river), riparian buffer strips, urban green-blue infrastructure (sponge city elements: bioswales, raingardens, permeable pavements, retention ponds), and coastal NbS (mangroves, salt marshes, dune restoration). NbS for floods sits at the intersection of "Ecosystem-based Adaptation" (EbA, CBD) and "Eco-DRR" (UNDRR/UNEP) — the distinction is largely framing rather than substance.

Vulnerable districts. Vulnerability in the IPCC AR6 sense is "the propensity or predisposition to be adversely affected" and is a function of sensitivity, exposure (treated separately in AR6), and adaptive capacity. In Indian operational practice, vulnerable districts are identified by (a) hazard layers (CWC flood-prone area maps, IMD rainfall climatology, NRSC/Bhuvan flood inundation atlases), (b) exposure layers (Census 2011 settlement density, agricultural land use, critical infrastructure), and (c) socio-economic vulnerability indices (the DST-Swiss Agency *Climate Vulnerability Assessment for the Indian Himalayan Region* methodology and its 2019/2021 nationwide extension by IIT-Mandi, IIT-Guwahati, and IISc, which produced district-level Vulnerability Index scores; CEEW's *Climate Vulnerability Index of Indian States and Districts* (2021); the NDMA District Disaster Management Plan (DDMP) vulnerability profiles). For this PhD, "vulnerable district" means a district that is in the top quartile on a composite *flood-specific* vulnerability index combining hazard recurrence (≥ 1 -in-5-year inundation), high exposure (population density on floodplain), and high social vulnerability (poverty headcount, SC/ST share, female-headed households, kuccha housing share, low literacy).

Measurable risk reduction. Following the IPCC formulation $\text{Risk} = f(\text{Hazard}, \text{Exposure}, \text{Vulnerability})$ and the Sendai Framework's seven global targets (with Targets A–D directly relevant: reduce mortality, affected people, economic loss, infrastructure damage), measurable risk reduction means an empirically demonstrable reduction in *one or more* of: (i) hazard intensity at the point of impact (e.g., peak flood depth, duration, velocity reduced through NbS attenuation), (ii) exposure (e.g., population and assets removed from harm's way through warning-triggered evacuation or floodplain zoning), (iii) vulnerability (e.g., improved housing, social protection, anticipatory action), or (iv) consequences (lives lost, injuries, livelihoods, displacement, infrastructure damage), measured against a credible counterfactual or baseline using designs robust to confounding.

Scalable model. Scalability has three nested meanings (after Hartmann and Linn, 2008; World Bank scaling-up framework). *Functional scaling*: doing the same thing in more places (replication). *Vertical scaling*: institutionalising in policy and budget so it persists. *Horizontal scaling*: integrating with adjacent sectors (health, agriculture, social protection). A truly scalable model satisfies all three, and explicitly designs for the *unit of replication* — the district — with an *enabling architecture* at the state and basin level.

These five definitions are the conceptual scaffolding for the rest of the brief and the dissertation.

Section 2. Global and Indian Evidence Review

This section is a critical synthesis, not a catalogue. The aim is to extract *transferable design knowledge* and *failure modes*, with a clear-eyed view of what is well-evidenced versus what is asserted by program reports.

2.1 Global overview — community flood EWS

Six exemplar cases are summarised below. They are chosen to span hazard types (riverine, flash, coastal, urban) and governance models.

(i) Bangladesh Cyclone Preparedness Programme (CPP) — extended to floods. Although originally a cyclone EWS established after the 1970 Bhola cyclone, the CPP — a joint Government of Bangladesh and Bangladesh Red Crescent Society programme operating ~76,000 trained volunteers in coastal upazilas — is the most-cited gold standard for last-mile community early warning. It works through tiered megaphones, hand-cranked sirens, flag signals, mosque announcements, and door-to-door volunteers, with explicit gender-balanced volunteer recruitment since the early 2010s. Effectiveness has been measured chiefly through cyclone mortality reduction: from ~500,000 deaths in 1970 to under 4,000 in Cyclone Sidr (2007) and under 200 in Amphan (2020), an order-of-magnitude reduction credited substantially (though not solely) to CPP and shelter infrastructure. For *floods*, the Flood Forecasting and Warning Centre (FFWC) under BWDB has progressively pushed forecasts to 5–10 day lead times, with FCAST-based community dissemination through CSOs (CDMP, RIMES). Lesson: large-scale, *paid-volunteer-led*, *gender-inclusive*, *redundant-channel* dissemination plus simple visual signalling is more robust than any single technological channel.

(ii) Mozambique – Búzi River community-based flood EWS. A river-basin-level pilot supported by UNDP, the Mozambique INGC, and the Red Cross, which combined upstream community-operated rainfall and water-level gauges (with radio handsets), a basin-level coordination centre, and downstream village-level response committees. Independent evaluations (post Cyclone Idai, 2019) found that villages with active CBEWS had measurably higher evacuation rates and lower asset losses than control villages, although the system was overwhelmed by the *magnitude* of Idai — a critical lesson on the limits of community EWS under low-probability, high-magnitude events without complementary structural and NbS measures.

(iii) Nepal – Karnali and Babai basins community flood EWS (Practical Action and partners). Upstream gauge readers (often women's groups) read staff gauges twice daily and communicate readings via mobile phones to a central operator, who issues colour-coded warnings (yellow / orange / red) downstream via SMS, sirens, FM radio, and door-to-door volunteers. Lead times of 3–8 hours have been documented for the Karnali. Effectiveness measures include (a) recorded false-alarm and missed-event rates against gauge data, (b) household evacuation rates, (c) asset pre-positioning behaviour. The Nepal model is widely cited as the most replicable in South Asia because it is *low-cost*, *open-protocol*, and *embedded in local government* (gaupalika and ward levels).

(iv) Cambodia / Mekong – community-based flood EWS (Mekong River Commission, ADPC). Combines basin-scale forecasting (MRC's Regional Flood Management and Mitigation Centre) with *village flood markers*, *colour-coded billboards*, and SMS cascades to commune leaders. Lessons: forecast skill at sub-basin scale was the limiting factor (not communication), and integration with *anticipatory cash transfers* (piloted with WFP and Start Network) has emerged as a high-value linkage.

(v) United States – flash-flood EWS in Appalachian and arid south-west catchments (NWS/NOAA + ALERT2 networks). Decentralised tipping-bucket rain gauges and stream gauges send data to local emergency-management offices using ALERT2 radio protocol; NWS issues Flash Flood Warnings. The American case is technologically rich but instructive on a different dimension: even with high-quality detection, *fatalities persist where last-mile response is weak* (drivers entering flooded roadways). Lesson: detection–dissemination–response is only as strong as the weakest link.

(vi) Europe – urban flash-flood EWS, e.g., Genoa (Italy) and Zurich (Switzerland). Dense urban rain-gauge and radar networks coupled with hydrodynamic urban models (e.g., MIKE URBAN, InfoWorks ICM) and operator-on-call decision protocols. Effectiveness measured in lead time at street level (target: 30–90 minutes) and false-alarm ratio. Lesson: in compact urban catchments, EWS lead time is hard-bounded by the catchment response time, so *reducing* the response time gap (faster public alerts, clearer evacuation routes, automated traffic control) is the operational lever.

Synthesis across global EWS cases. What recurs: (a) *redundancy* of channels (radio + SMS + sirens + door-to-door), (b) *trusted local intermediaries* (volunteers, faith leaders, women's collectives), (c) *forecast skill* matters but is rarely the binding constraint, (d) *response infrastructure* (shelters, evacuation routes, anticipatory cash) is decisive, and (e) *gender-balanced volunteering* materially raises last-mile efficacy.

2.2 Global overview — nature-based flood protection

(i) Netherlands – Room for the River (Ruimte voor de Rivier, 2007–2019). The most-studied European NbS-style intervention (though many Dutch engineers prefer "spatial water management" as the framing). Across 30+ sites along the Rhine, Waal, IJssel, and Meuse, the program lowered floodplains, relocated dikes inland, created bypass channels, and restored floodplain forests, achieving the design discharge target (16,000 m³/s at Lobith) while delivering ecological co-benefits. Effectiveness measured through (a) modelled and observed reduction in design water levels (typically 30–50 cm at key gauges), (b) ecological indicators (riparian vegetation, fish connectivity), (c) land-use multifunctionality. Lesson: NbS at scale requires *long political horizons* (decades), *strong spatial planning powers*, and *multi-objective valuation* that goes beyond flood safety.

(ii) United States – Mississippi River and Louisiana coastal NbS. Sediment diversions, marsh creation, oyster reef restoration under the Louisiana Coastal Master Plan. Effectiveness measured through hydrodynamic-morphodynamic modelling (e.g., Delft3D, ADCIRC+SWAN) projecting marsh accretion and storm-surge attenuation; ex-post measurement is harder due to compounding processes (subsidence, sediment starvation). Lesson: NbS at coastal scales has long lead times before measurable risk reduction emerges; interim "no-regret" co-benefits (fisheries, carbon) sustain political will.

(iii) United Kingdom – Natural Flood Management (NFM), e.g., Pickering, Holnicote, Stroud Frome. Upstream leaky woody dams, wetland creation, peatland restoration, soil-aeration, hedgerow planting. Pickering (North Yorkshire) is a textbook case where modelled and limited observational evidence suggested ~15–20% reduction in peak flow for moderate events; uncertainty is acknowledged for extreme events. UK Defra and Environment Agency have funded a national NFM evidence-base that *honestly reports uncertainty*. Lesson: NbS effectiveness is event-magnitude-dependent; benefits are clearest for small-to-moderate floods and attenuate (in relative terms) for extreme events.

(iv) China – Sponge City Programme (2015 onward; pilots in 30+ cities incl. Wuhan, Zhenjiang, Xiamen). Urban NbS at scale: bioswales, permeable pavements, retention ponds, green roofs targeting 70% rainfall retention for design events. Mixed evaluation evidence: where well-implemented (retrofits in new districts), measurable reductions in surface ponding and combined-sewer overflows; where poorly maintained or undersized for extreme events (e.g., Zhengzhou 2021), failure was severe. Lesson: NbS in urban contexts must be designed for *multi-scale events* with explicit residual-risk planning.

(v) Vietnam / Philippines / Bangladesh – mangrove and bioshield restoration (post-2004 tsunami, post-Haiyan, Sundarbans). Empirical literature (e.g., Menendez et al., *Sci Reports* 2020) estimates mangroves reduce annual flooding for 18+ million people globally. Dose-response: a 100 m mangrove belt reduces wave height by 13–66% depending on density and tree age. Lesson: mangroves are a high-evidence NbS for *coastal* flood and storm-surge attenuation; they require tenure

security and livelihoods-linked stewardship to persist.

(vi) Africa – Tana River basin (Kenya) and Volta basin – upstream catchment treatment. Combinations of soil-and-moisture conservation, terracing, agroforestry, rangeland management measured through SWAT-modelled and observed reductions in sediment yield and peak runoff. Effectiveness is moderate and slow-emerging; co-benefits for soil fertility, livelihoods, and carbon are typically more measurable than the flood-specific signal. Lesson: in dryland basins, NbS framing benefits from foregrounding multifunctionality.

Synthesis across global NbS cases. What recurs: (a) effectiveness is *event-magnitude-dependent* (best for small-to-moderate, partial for extreme), (b) the *measurement problem* is harder than for EWS — attribution of hydrological effects requires long monitoring records or paired-catchment designs, (c) co-benefits (biodiversity, carbon, livelihoods) are often the political glue, (d) maintenance and tenure are decisive for persistence, and (e) hybrid grey–green designs typically outperform purely green or purely grey approaches in cost-benefit terms.

2.3 India and South Asia focus — selected documented cases

(i) Assam – Brahmaputra basin community flood EWS (Aaranyak / SEEDS / RIMES / NDMA pilots in Lakhimpur, Dhemaji, Majuli, Goalpara). Volunteer gauge readers along tributaries (e.g., Jiadhal, Jia Bharali, Subansiri), SMS cascades, FM radio, and women-led village committees ("village resilience committees" or *gaon-rakhya samiti*) have been piloted since the mid-2010s. Documented strengths: improved local recognition of warning thresholds, reduced livestock losses (a major asset class in Assam), embedded women's leadership in some sites. Documented weaknesses: discontinuity when project funding ended, fragmented data flow to ASDMA, and patchy integration with CWC's own forecasts. The 2024 Assam floods exposed persistent last-mile gaps even where pilots had operated.

(ii) Bihar – Kosi basin EWS and post-2008 flood reconstruction. The 2008 Kosi breach (Kusaha) killed ~500 and displaced ~3 million. Subsequent Bihar State Disaster Management Authority (BSDMA) programs, with World Bank Bihar Kosi Flood Recovery Project support, deployed embankment monitoring, community task forces, and *Aapda Mitra*-style volunteer cadres. The CWC's North Bihar flood forecasting has improved (Birpur, Baltara, Kursela gauges). Yet the binding constraint identified in independent reviews is *transboundary data sharing with Nepal* (limited real-time exchange of upstream Kosi/Bagmati/Kamla discharges), which constrains lead time. Lesson: in transboundary basins, lead time is bounded by data diplomacy as much as by technology.

(iii) Odisha – cyclone EWS and coastal NbS (post-1999 Super Cyclone trajectory). Odisha's transformation from the 1999 cyclone (~10,000 deaths) to Cyclone Phailin (2013, <50 deaths) and Fani (2019, ~64 deaths) is the most-cited Indian success story for *integrated* EWS + shelter infrastructure + evacuation protocols + (more recently) mangrove restoration in Bhitarkanika and Mahanadi delta. The Odisha State Disaster Management Authority (OSDMA) operates SMS-based mass alerts, sirens, and a state-funded volunteer cadre, with demonstrated equity gains (Persons with Disabilities included in evacuation planning). For floods specifically (as distinct from cyclones), basin-level community EWS in the Mahanadi remains less mature.

(iv) Uttarakhand – flash-flood and GLOF EWS (post 2013 Kedarnath, 2021 Chamoli). Pilots by Uttarakhand SDMA, NDMA, IIT-Roorkee, DRDO-DGRE, and the Central Water Commission combine glacial-lake monitoring, sirens along Char Dham routes, and pilgrim-targeted SMS. Lessons centre on (a) the inadequacy of static EWS in non-stationary cryosphere settings, (b) the criticality of *route-based* warning for a transient pilgrim population, and (c) the limits of structural responses alone.

(v) Chennai / Mumbai – urban flood EWS. Chennai's C-FLOWS (post-2015 deluge) and Mumbai's I-FLOWS (IITM-MoES-BMC) are operational urban flood forecasting platforms providing 6–72 hour warnings. Strengths: high-resolution rainfall now-casting and hydraulic modelling. Weaknesses: limited last-mile dissemination beyond government SMS systems and uneven *uptake by vulnerable*

settlements (informal housing, fishing communities). Inclusion-weighted EWS performance is largely unmeasured.

(vi) Sikkim and the Eastern Himalayas – GLOF EWS. Post South Lhonak Lake event (October 2023), accelerated investment in glacier-lake instrumentation. The case underscores how *new hazard regimes* (cascading cryospheric–riverine events) demand new EWS architectures and force a re-think of "vulnerable district" beyond plains/coastal floods.

On NbS in India — selected documented cases.

- *Watershed development under NMSA / IWMP / MGNREGS.* Decades of soil-and-moisture conservation, contour bunding, check dams, farm ponds across Maharashtra, Madhya Pradesh, Rajasthan, Andhra Pradesh, Karnataka, Telangana. Evaluations (NCAP-NIPFP, IWMI, IFPRI, ICRISAT, CSE) show high *agronomic* returns and moderate but measurable hydrological effects (groundwater recharge, baseflow). The flood-specific signal is rarely isolated and remains a research gap.
- *Wetlands and tank cascade restoration.* Tamil Nadu's *eri* (tank) system rehabilitation, Telangana's *Mission Kakatiya*, West Bengal's East Kolkata Wetlands. Hydrological co-benefits for urban flood attenuation are documented qualitatively; quantitative attribution is partial.
- *Mangrove restoration in Sundarbans, Bhitarkanika, Pichavaram, Gulf of Kachchh.* The MISHTI scheme (2023, Union Budget) and state-level programs. Strong evidence base for storm-surge attenuation; flood (riverine) co-benefits are smaller and basin-specific.
- *Namami Gange and riparian buffer / wetland-corridor work along the Ganga.* Mixed implementation; flood-risk-reduction co-benefits not yet rigorously quantified at sub-basin scale.
- *Urban green-blue infrastructure pilots.* Bengaluru lake rejuvenation, Hyderabad's nala restoration, Chennai's Pallikaranai marsh. Promising but uneven; generally under-monitored for flood outcomes.

How are vulnerable districts currently identified in India? The principal national-level instruments are: (a) NDMA's hazard atlases and District Disaster Management Plans; (b) the DST/SDC-funded *Climate Vulnerability Assessment* (Sharma et al., 2019; nationwide extension 2021) which provides a district-level Vulnerability Index using IPCC AR5 framing (sensitivity + adaptive capacity); (c) CEEW's *Climate Vulnerability Index* (2021) using the IPCC AR5 framing across 640+ districts and 8 climate hazards; (d) NRSC/Bhuvan flood inundation atlases derived from satellite archives; (e) state-specific tools (Assam SDMA's basin-scale hazard mapping, Odisha's multi-hazard index). **Gaps:** integration of *flood-specific* vulnerability with EWS coverage and NbS *opportunity* layers; absence of *equity-disaggregated* vulnerability layers (gender, caste, tenure); weak linkage between district vulnerability scores and DDMP budgeting and project pipelines.

2.4 Synthesised design principles and failure modes

Design principles for effective community flood EWS (extracted across cases):

1. *Co-produced risk knowledge* with communities, not delivered to them.
2. *Redundant communication channels* (at least three: e.g., SMS + siren + door-to-door volunteer + community radio).
3. *Tiered, colour-coded* warning thresholds tied to local visual cues (river marks, gauge boards).
4. *Trusted local intermediaries*, gender-balanced and including marginalised groups.
5. *Anticipatory action protocols* (pre-positioned cash, livestock-shelter pairing, evacuation routes drilled annually).
6. *Forecast-to-warning translation rules* documented and rehearsed (no ad-hoc calls during events).

7. *Equity audit* of who actually receives and acts on warnings.
8. *Embedded in district disaster management plans* with budget lines, not project-financed only.
9. *Open data and open protocols* to allow horizontal scaling.
10. *Annual exercises* and after-action reviews tied to indicator dashboards.

Design principles for effective NbS for floods:

1. *Watershed/basin scale logic*, not parcel-by-parcel logic.
2. *Hybrid grey–green* designs where extreme-event residual risk persists.
3. *Tenure and rights clarity* before intervention (especially for forests, wetlands, mangroves).
4. *Multi-objective valuation* (flood + water + livelihoods + biodiversity + carbon).
5. *Community stewardship arrangements* with maintenance budgets.
6. *Modelling–monitoring–adaptive management* loops (e.g., paired-catchment monitoring).
7. *Climate-change-adjusted design* (non-stationary hydrology).
8. *Spatial planning integration* (zoning, buffer rules, master plans).
9. *Sequencing*: NbS that mature slowly need interim risk-management measures.
10. *Cost–benefit honesty* about residual extreme-event risk.

Common failure modes (across both EWS and NbS):

- *Technical*: forecast-to-warning translation gap; data discontinuity when project ends; un-maintained gauges; un-maintained NbS structures; under-design for extreme events.
- *Social*: warning reach but no warning *equity*; absence of women and marginalised groups in design; mistrust of government channels; livestock and asset evacuation neglected.
- *Institutional*: weak DDMP integration; unclear roles between SDMA, panchayat, line departments; absence of recurrent budget lines; staff turnover.
- *Financial*: project-only funding with no exit financing strategy; CSR fragmentation; absence of cost-benefit narratives that win recurrent fiscal allocations.

Section 3. Conceptual and Analytical Framework

3.1 The framework in prose

The framework is built on the IPCC AR6 risk identity, **Risk = f(Hazard, Exposure, Vulnerability, Response)**, with response itself disaggregated into anticipatory and reactive components. Three nested layers are articulated.

Layer 1 — Hazard formation. Climate drivers (monsoon dynamics, extreme rainfall, snow/glacier melt, cyclonic systems) interact with catchment characteristics (land use, soil, drainage density, channel geometry, sediment dynamics) to produce a hydrological hazard signature: rainfall intensity-duration-frequency, river discharge hydrograph, urban drainage congestion. *NbS interventions* operate primarily on Layer 1 by modifying catchment response: upstream catchment treatment reduces and delays peak runoff; floodplain reconnection reduces peak stage; wetlands attenuate; mangroves dissipate surge energy. Layer 1 is where hydrological *attenuation* — the physical reduction of hazard intensity at the point of impact — is achieved.

Layer 2 — Exposure and vulnerability. Settlements, livelihoods, infrastructure, ecosystems, and people are exposed in spatially explicit ways (floodplain occupation, asset density, critical infrastructure footprints). Vulnerability is shaped by socio-economic conditions (poverty, housing quality, livelihoods, social protection), demographic factors (gender, age, caste, disability), and information access (literacy, language, phone ownership, media exposure). *NbS* affects exposure indirectly via spatial planning (e.g., floodplain zoning attached to *NbS* schemes) and rarely affects vulnerability directly. *EWS* affects exposure through warning-triggered evacuation and asset pre-positioning, and affects vulnerability through preparedness and anticipatory cash transfers.

Layer 3 — Response and consequence. The four *EWS* components — risk knowledge, monitoring/forecasting, dissemination/communication, response capability — sit in this layer and convert hazard and exposure into mitigated or unmitigated consequences. Consequences are measured in lives, livelihoods, assets, ecosystems, and recovery time. The *governance and financing layer* sits as a cross-cutting enabler, determining whether the technical and social components are sustained.

The **theory of change** is that an *integrated EWS + NbS + institutional strengthening* intervention produces measurable risk reduction through four co-acting mechanisms: (M1) *hazard attenuation* via *NbS* at watershed scale; (M2) *exposure reduction* via warning-triggered protective action; (M3) *vulnerability reduction* via preparedness, anticipatory finance, and inclusive design; (M4) *response amplification* via institutional readiness and infrastructure. The integrated model dominates the "either/or" alternatives because the four mechanisms are *complementary* — *NbS* handles small-to-moderate events well but leaves residual extreme risk that *EWS+evacuation* handles; *EWS* handles tail events but does not change the underlying hazard intensity.

3.2 The framework as a diagram

The diagram has six boxes and four feedback loops.

Boxes (left to right and top to bottom):

1. **Climate drivers** (rainfall, river discharge, sea level, cryosphere) — top-left.
2. **Catchment / coastal system** (land use, soils, channel, ecosystems) — receives arrows from Box 1; this is where *NbS* operates.
3. **Hazard at impact point** (depth, duration, velocity, extent) — receives arrows from Box 2.
4. **Exposed and vulnerable system** (settlements, people, livelihoods, infrastructure) — sits parallel to Box 3.

5. **EWS chain** (risk knowledge → monitoring → dissemination → response capability) — bridges Boxes 3 and 4.
6. **Consequence** (lives, livelihoods, assets, displacement, ecosystems) — bottom-right, the outcome variable.

A cross-cutting **governance and financing layer** (DDMA, SDMA, panchayat, line departments, CSO, community institutions, funders) underlies all six boxes.

Arrows (causal pathways):

- 1 → 2 → 3: climate-to-hazard.
- 2 ← *NbS interventions*: NbS modifies the catchment response.
- 3 → 5 → 4: hazard triggers EWS chain which triggers protective action in the exposed system.
- 4 → 6: the exposed-vulnerable system absorbs the hazard and produces the consequence.
- 3 → 6 (residual): unmitigated hazard directly produces consequence (where EWS or NbS underperform).

Feedback loops:

- F1 (*learning loop*): consequences feed back to risk knowledge (after-action reviews → updated thresholds, maps).
- F2 (*ecological loop*): NbS interventions yield co-benefits (groundwater, biodiversity, carbon, livelihoods) that feed back as community willingness-to-maintain.
- F3 (*institutional loop*): performance indicators feed back to budget lines and DDMP updates, locking in vertical scaling.
- F4 (*equity loop*): equity audits feed back to redesign of channels and intermediaries, closing the warning-equity gap.

Where measurable risk reduction occurs in the framework. Four loci are identifiable:

- *Reduced hazard intensity at the impact point* (Box 3 metrics: peak depth, duration, extent), attributable mainly to NbS (Box 2 modifications).
- *Reduced exposure during the event* (Box 4 metric: people and assets in harm's way at peak), attributable mainly to EWS-triggered protective action.
- *Reduced vulnerability* (Box 4 metric: housing quality, social protection coverage, anticipatory finance), attributable to preparedness investments.
- *Reduced consequence* (Box 6 metrics: lives, livelihoods, assets), attributable to the joint action of all the above.

This decomposition is essential because *attribution* in evaluation requires knowing which mechanism produced which effect — a recurring shortcoming in current monitoring frameworks, which conflate output (e.g., "X warnings issued") with outcome (e.g., "Y people protected").

3.3 Three alternative theories of change

ToC-A: Community EWS only. *Assumptions*: forecasts are sufficiently skilful; communication channels reach the most vulnerable; communities have the means and time to act on warnings; institutions sustain the system. *Activities*: install gauges, train volunteers, deploy SMS/sirens, run drills, integrate with DDMP. *Intermediate outcomes*: warning lead time achieved; warning reach (% of population alerted); response rate (% of alerted who act); pre-positioning of assets and livestock. *Final*

outcomes: reduced mortality and injuries; reduced asset losses; faster recovery. *Limitations*: no change in hazard intensity; tail risk for extreme events with insufficient lead time; equity gap unaddressed if last-mile is weak; behavioural fatigue under repeated false alarms.

ToC-B: NbS only. *Assumptions*: catchment-scale interventions can be implemented and maintained; hydrological response is favourable; tenure and rights are clarified; co-benefits sustain political and community will. *Activities*: catchment treatment, wetland restoration, floodplain reconnection, mangrove planting, riparian buffers, urban green-blue retrofits. *Intermediate outcomes*: increased infiltration and storage capacity; delayed and reduced peak runoff; restored ecosystem health; livelihoods co-benefits. *Final outcomes*: reduced hazard intensity for small-to-moderate events; reduced damages from frequent floods; ecosystem co-benefits (biodiversity, carbon, water security). *Limitations*: residual extreme-event risk remains uncovered; benefits emerge slowly (years to decades); attribution to flood reduction is hard to prove; vulnerable to maintenance failure.

ToC-C: Integrated EWS + NbS + institutional strengthening (the proposed ILRM). *Assumptions*: institutional leadership at district level can be sustained; finance can be blended; communities are partners; data and protocols are open. *Activities*: all of A and B, plus DDMP integration, recurrent state budget lines, equity-disaggregated indicator system, anticipatory cash transfers, multi-stakeholder governance platform. *Intermediate outcomes*: NbS hydrological co-benefits; EWS lead time and reach with equity; institutional readiness; anticipatory action triggered. *Final outcomes*: reduced hazard intensity (small-moderate events), reduced exposure and consequences (all events including extreme), strengthened adaptive capacity, persistent risk reduction across decades. *Why it dominates*: complementarity of mechanisms; resilience to single-component failure; alignment with funder logic (multiple SDGs, both adaptation and mitigation co-benefits, both DRR and ecosystem outcomes).

The PhD inquiry can use this triad as the core *comparative* logic — RQ1 is essentially a test of whether the integrated model out-performs A or B.

Section 4. Measurable Risk Reduction — Indicators and Methods

4.1 Indicator set (multi-tier)

A funder-credible indicator set must span four tiers: risk-reduction outcomes, system performance, social equity, and ecosystem co-benefits. The set below is calibrated to be *measurable with realistic data, aligned with Sendai Targets A–D, and compatible with GCF/AF/World Bank logframes*.

Tier 1 — Risk reduction outcomes.

- *Indicator 1.1: Mortality avoided.* Definition: deaths attributable to flood events per 100,000 exposed population, compared to a counterfactual (pre-intervention or control district) baseline. Units: deaths/100,000. Sources: SDMA SitReps, NDMA disaster database, district health records, NCRB cross-checks. Frequency: per event + annual aggregate. Pros: directly maps to Sendai Target A and is the highest-salience outcome for funders. Cons: low-frequency events make statistical detection hard; data quality is uneven; attribution requires controls.
- *Indicator 1.2: Injuries and morbidity.* Hospitalisations and reported injuries per 100,000 in the affected population. Sources: district hospital records, ASHA/ANM reports. Cons: under-reporting bias.
- *Indicator 1.3: Affected population.* Sendai Target B aligned. Census + SDMA data, cross-checked with satellite-derived inundation × population layers (e.g., WorldPop, HRSL).
- *Indicator 1.4: Economic losses (direct).* Damage to housing, agriculture, livestock, infrastructure, in INR per affected household and aggregate. Sources: SDMA damage assessments, agricultural department crop loss reports, household survey panels. Frequency: per event + annual. Pros: enables benefit-cost analysis. Cons: valuation is contested (replacement cost vs. depreciated value vs. willingness-to-pay).
- *Indicator 1.5: Crop area inundated and crop loss.* Hectares × loss percentage; satellite + ground truth.
- *Indicator 1.6: Displacement-days.* Number of person-days spent in shelters or displaced from primary residence. Captures duration of disruption, not just incidence.
- *Indicator 1.7: Reduction in peak flood depth and extent at impact points.* Modelled and observed (Sentinel-1 SAR pre/during/post) inundation maps; with-and-without intervention scenarios. Hydrological attribution metric — most directly linked to NbS effectiveness.

Tier 2 — System performance.

- *Indicator 2.1: Forecast skill at the relevant lead time.* Standard verification metrics — Probability of Detection (POD), False Alarm Ratio (FAR), Critical Success Index (CSI), Brier Skill Score, RMSE — computed against observations. Sources: CWC/IMD products vs. gauge data.
- *Indicator 2.2: Warning lead time.* Time from warning issuance to flood arrival at the impact point. Hours. Median and distribution.
- *Indicator 2.3: Warning coverage.* Share of at-risk population reached by at least one warning channel. From household surveys and channel logs.
- *Indicator 2.4: Warning reach with comprehension.* Share of at-risk population that received the warning *and* understood it correctly. Survey-based.
- *Indicator 2.5: Warning-to-action rate.* Share of warned households that took at least one prescribed protective action. Distinguishes "informed" from "protected".
- *Indicator 2.6: Response time of district machinery.* Hours from warning to deployment of SDRF/NDRF, opening of shelters, activation of anticipatory cash.

- *Indicator 2.7: System uptime.* Share of monsoon season during which all gauges, channels, and protocols are fully functional. A maintenance discipline indicator.

Tier 3 — Social equity.

- *Indicator 3.1: Equity-disaggregated warning reach.* Tier-2 indicators broken down by gender, caste (SC/ST/OBC/General), tenure (owner/tenant/landless), housing type (kuccha/pucca), disability status, and language. The *equity gap ratio* (most-vs-least reached subgroup) is a single-number summary.
- *Indicator 3.2: Inclusion in design and governance.* Share of women, SC/ST, and PwD in volunteer cadres, village resilience committees, DDMA consultations.
- *Indicator 3.3: Equity-weighted avoided losses.* Avoided losses weighted by household poverty status (e.g., MPI-weighted).
- *Indicator 3.4: Anticipatory cash transfer coverage.* Share of eligible vulnerable households receiving pre-event cash, with timing relative to event onset.

Tier 4 — Ecosystem co-benefits.

- *Indicator 4.1: NbS area treated and condition.* Hectares of catchment treated, wetland restored, mangrove planted/regenerated, with condition class (vegetation cover, density, species composition).
- *Indicator 4.2: Hydrological co-benefits.* Groundwater level change (CGWB observation wells), baseflow index, sediment yield reduction (where measurable).
- *Indicator 4.3: Biodiversity indicators.* Species richness in restored sites; presence/absence of indicator species; bird counts.
- *Indicator 4.4: Carbon co-benefits.* Estimated above- and below-ground carbon stocks; for mangroves, blue carbon estimates.
- *Indicator 4.5: Water quality.* Standard parameters (DO, BOD, turbidity, faecal coliform) where wetland and riparian NbS is implemented.
- *Indicator 4.6: Livelihood co-benefits.* Income from NbS-linked activities (NTFP, fisheries, agroforestry); MGNREGS person-days created.

Pros and cons across the indicator set. The risk-reduction outcomes (Tier 1) are the most funder-salient but the hardest to measure causally because flood events are episodic and confounded by hydrology. System-performance indicators (Tier 2) are tractable, regular, and useful for management. Equity indicators (Tier 3) are critical and currently neglected; they require small but dedicated household-survey effort. Ecosystem co-benefits (Tier 4) are a competitive advantage for funders aligned with biodiversity and carbon agendas (GCF, GEF, AF, CBD, UNFCCC NDC).

A practical tip: report Tier 1 with explicit uncertainty bounds and counterfactuals, Tier 2 monthly, Tier 3 annually with disaggregation, Tier 4 annually with seasonal sampling.

4.2 Mixed-methods evaluation designs realistic for a PhD scholar in India

Quantitative designs.

- *Before–after comparison within the pilot district(s).* Simple, but exposed to confounding by climate variability between periods. Strengthen with multi-year baselines and statistical controls.
- *Difference-in-differences (DiD) across pilot and control districts.* Identifies the intervention effect under the parallel-trends assumption. Recommended where two contrasting districts (one intervention, one control) can be selected on observables.

- *Synthetic control method.* Constructs a weighted "synthetic" control district from a donor pool. Particularly suitable for single-district interventions; under-used in DRR research and offers a publishable methodological contribution.
- *Hydraulic / hydrological modelling with-and-without scenarios.* HEC-RAS, HEC-HMS, LISFLOOD-FP, MIKE Flood, SWAT, CaMa-Flood — modelled pre/post NbS scenarios for design events. Realistic for a PhD scholar with open-source tools and DEM (SRTM 30 m, ALOS-PALSAR 12.5 m, CartoDEM 30 m where accessible).
- *Remote-sensing flood extent mapping.* Sentinel-1 SAR time series (10 m, 6–12 day revisit), Sentinel-2 optical, MODIS for large events; flood frequency maps over 5–10 years using Google Earth Engine — accessible, free, and highly publishable.
- *Survey-based exposure and vulnerability indices.* Household panel surveys (sample 600–1200 households across treatment and control villages) with structured questionnaires aligned to SVI (Social Vulnerability Index) and multidimensional poverty.

Qualitative designs.

- *Participatory hazard and resource mapping.* PRA tools at village level; produces both data and community ownership.
- *Focus group discussions* segmented by gender, caste, age, occupation; standard analytical software (NVivo / open-source alternatives like Taguette).
- *Institutional analysis* (semi-structured interviews with DDMA, SDMA, panchayat, CWC, IMD, line departments, CSOs). Use Ostrom's IAD framework or the Sendai governance lens.
- *Ethnographic immersion* — even short (4–8 weeks) in one or two villages — yields decisive insight into the warning-to-action gap, household decision-making, gendered evacuation dynamics, and livestock-asset trade-offs.
- *After-action reviews* of flood events during the field period — one of the most data-rich opportunities for a flood researcher.

What is realistically feasible for a part-time PhD in India. A lone scholar cannot run a full RCT, and frankly, RCTs are usually inappropriate for community DRR. A *quasi-experimental DiD or synthetic control* across two districts, *coupled with* hydraulic modelling for hazard attribution, *coupled with* mixed-methods qualitative work for mechanism evidence, is rigorous and tractable. Partnerships are decisive: SDMA / DDMA for data and access; one credible research institute (IIT, NIH-Roorkee, IIRS, NIDM, NIAS, CEEW, ATREE) for computation and credibility; one CSO (SEEDS, Aaranyak, Practical Action, ASAR, Goonj, Welthungerhilfe) for community access; and one panchayat-level entity for ground operations. A part-time scholar should *negotiate co-supervision* with one academic and one practitioner mentor.

Dealing with uncertainty. Be explicit. Use Bayesian or bootstrap confidence intervals where sample sizes are small. Report effect sizes with credible intervals. State the magnitude of confounders. Use *triangulation* — quantitative DiD plus qualitative process-tracing plus modelled hazard attribution — to make a defensible inference even when each strand has limitations.

Section 5. Scalable Implementation and Funding Models

5.1 What "scalable" realistically means in the Indian district context

Three scales are operationally meaningful for India.

- *Sub-state cluster scale (5–10 districts within a basin or state).* Achievable in 3–5 years through a state SDMA-led program with one or two implementing partners and a research-evaluation arm. Most defensible target for a PhD-anchored model.
- *State scale (20–40 districts).* Achievable in 5–10 years if the model is institutionalised in DDMP guidelines, integrated with MGNREGS and the State Action Plan on Climate Change (SAPCC), and embedded in recurring state Finance Commission allocations and the National Disaster Risk Management Fund (NDRMF) / SDRMF lines.
- *National basin-scale (e.g., Brahmaputra, Ganga, Mahanadi, Krishna).* Requires inter-state coordination through CWC, the National Disaster Management Authority, and a basin-level platform. 10–15 year horizon.

The mechanisms for scaling are: (a) *standardisation* of protocols, hardware, training curricula, and indicator frameworks; (b) *open data and open APIs* so that any district can plug in; (c) *recurring public budget lines* (SDRMF, MGNREGS, NAFCC, central scheme convergence); (d) *capacity development* networks through NIDM, ATIs, SIRDs; (e) *blended finance* combining grants (MoEFCC, MoJS, World Bank), philanthropic CSR (top corporate spenders on climate/DRR), and concessional loans (AIIB, ADB, World Bank); (f) *anticipatory finance* mechanisms (IFRC's anticipatory action protocols, WFP / Start Network experience, contingent disaster financing).

5.2 Three to four scalable implementation models

Model A — State-led DDMP-integrated program. *Architecture:* SDMA leads, DDMA implements, a state-level technical agency (e.g., a state remote-sensing centre, a state university) provides the data backbone. Community EWS and NbS interventions are integrated as standing line items in District Disaster Management Plans and the state's annual disaster preparedness budget. *Funding:* SDRMF + MGNREGS (for catchment and wetland works) + NAFCC + state plan funds + central scheme convergence (Jal Shakti Abhiyan, Namami Gange, MISHTI, Atal Bhujal). Ticket size: INR 5–25 crore per district per year recurrent, plus 50–150 crore one-time capital for major NbS works at sub-basin scale. *Roles:* SDMA — protocols, monitoring, technical leadership; DDMA — operations, last-mile, community interface; line departments — sectoral integration; community institutions (panchayats, women's collectives, *jal-doot / aapda mitra* cadres) — last-mile delivery. *Risks:* political turnover; bureaucratic inertia; weak inter-departmental coordination; under-funded recurring O&M. *Enablers:* strong SDMA leadership; chief secretary's office sponsorship; state climate cell embedment; alignment with SAPCC. *Funder fit:* World Bank R-DRIP, AIIB DRR, GCF (national-level access through NABARD).

Model B — NGO–government partnership with community EWS + NbS micro-projects. *Architecture:* A consortium of established NGOs (e.g., one technical lead for EWS, one ecological lead for NbS, one community lead for inclusion) partners with the SDMA/DDMA under an MoU. Pilots are co-designed with panchayats and ward-level institutions. *Funding:* NAFCC + bilateral (FCDO, GIZ, USAID, JICA, AFD) + climate philanthropy (Climateworks, Bezos Earth Fund, IKEA Foundation, MacArthur, Tata Trusts, Azim Premji) + CSR. Ticket size: INR 3–10 crore per district over 3–5 years for pilots; INR 25–60 crore for cluster expansion. *Roles:* NGO consortium — design, training, monitoring; SDMA/DDMA — endorsement, data access, scaling pathway; community institutions — operations. *Risks:* project-funding cliffs; weak vertical scaling without state ownership; staff turnover in NGOs. *Enablers:* clear handover-to-state protocols from year 1; research-evaluation arm publishing peer-reviewed evidence to convince state finance. *Funder fit:* bilateral, GCF readiness, philanthropic.

Model C — Tech + data platform with local implementation partners. *Architecture:* A technology and data platform provider (private, social-enterprise, or PRI like IIT/IIT spinout) provides forecasting, dashboards, dissemination APIs, and analytics across multiple districts. Local implementation partners (NGOs, panchayats, SHG federations) deliver community EWS and NbS on the ground. *Funding:* Hybrid: central/state procurement (Smart Cities, AMRUT 2.0 for urban; SDMA tech budgets for rural); CSR; impact investment for the platform. *Roles:* Platform — forecasting, data, dashboards, training content; local partners — community processes; SDMA/DDMA — governance and last-mile authority. *Risks:* vendor lock-in; data sovereignty concerns; misalignment between platform's commercial logic and community's stewardship logic. *Enablers:* open standards; transparent procurement; data-sharing agreements; community-owned data layers. *Funder fit:* Smart Cities, AMRUT, MeitY innovation funds, CSR climate-tech, impact investors.

Model D — Basin-scale public-philanthropic-multilateral compact. *Architecture:* A basin-scale platform (e.g., Brahmaputra, Mahanadi, Kosi) under CWC/MoJS leadership with state SDMA, multilateral co-financiers, and accredited entities (NABARD as GCF NIE, SIDBI, IREDA), plus a basin secretariat. *Funding:* GCF + AIIB + World Bank + AF + central + state + philanthropy. Ticket size: USD 100–500 million over 7–10 years. *Roles:* basin secretariat coordinates planning; state SDMA implement; CWC/IMD provide forecasting backbone; line departments execute NbS at scale. *Risks:* governance complexity; transboundary frictions (where applicable); long mobilisation time. *Enablers:* an accredited entity (NABARD) with GCF capacity; political backing at union level; strong M&E. *Funder fit:* GCF flagship; AF; World Bank programmatic; AIIB.

5.3 What makes a proposal attractive to climate funders

Across GCF, AF, World Bank, AIIB, and major foundations the recurring expectations are remarkably similar. A proposal lands well when it shows:

1. *A theory of change written in the funder's language* (results framework, impact–outcome–output–activity logic, with explicit risk-reduction logframe).
2. *Measurable indicators* aligned to recognised frameworks (Sendai Targets A–D; GCF result areas; SDG indicators 11.5, 13.1, 13.2, 13.3; CBD targets where ecosystem co-benefits exist).
3. *An evidence base* — either pilots' published results or rigorous analogous-project evidence. Peer-reviewed papers and independent evaluations are decisive.
4. *Cost–benefit or cost-effectiveness narrative* with avoided-loss accounting, ideally probabilistic (expected annual loss avoided, EAL%) and climate-change-adjusted.
5. *Co-benefits* — adaptation + mitigation (for GCF in particular), biodiversity, livelihoods, gender, social protection. The integrated EWS+NbS model scores well here.
6. *Gender and inclusion strategy*, with budget lines and indicators (essential for GCF and AF; increasingly for World Bank and bilaterals).
7. *Sustainability and exit strategy*: how the model continues after the grant ends; institutional embedment in state budgets.
8. *Scalability pathway*: replication architecture with unit costs and timelines.
9. *Transparent risk register*: climate, political, fiduciary, social risks with mitigation measures.
10. *Strong implementing entity track record*: most funders judge the entity as much as the project.

For a PhD-anchored proposal, the academic rigour is itself an asset — funders increasingly reward proposals with embedded MERL (monitoring, evaluation, research, learning) arms led by credible academic partners.

Section 6. India-Focused PhD Research and Pilot Designs

This section translates the framework into concrete PhD designs. The premise is one part-time scholar with a practitioner role, working in 1–2 vulnerable Indian districts over 3–4 years, with realistic data and partnership access.

6.1 Three candidate designs

Design 1 — Comparative case study of two contrasting districts.

District selection: District X = a Brahmaputra-basin district with relatively mature EWS pilots (e.g., Dhemaji, Lakhimpur, or Majuli, Assam). District Y = a Kosi-basin district with high recurrence but weaker last-mile (e.g., Supaul, Madhubani, Saharsa, Bihar).

Main RQs and hypotheses: Does the more mature EWS+NbS configuration in District X produce measurably greater risk reduction, and through which mechanisms, compared to District Y? H1: warning lead time and coverage are higher in X; H2: equity-disaggregated reach gap is smaller in X; H3: avoided losses per capita are higher in X for events of comparable magnitude.

Data required: hydromet (CWC discharge, IMD AWS, gridded rainfall); satellite (Sentinel-1 SAR, GPM-IMERG, MODIS); socio-economic (Census 2011, SECC, NFHS-5 for district characteristics); institutional (DDMP, SDMA logs, after-action reports); household surveys ($n \approx 600$ –1000 across treatment and comparison villages); key informant interviews.

Methods: DiD or synthetic control on event-level losses; hydraulic modelling for hazard attribution; mixed-methods qualitative analysis for mechanism evidence; equity audit.

Ethics and practicalities: IRB approval (most Indian universities now require it); informed consent; sensitivity in disaster contexts; partnerships with SDMA and CSOs for access; do-no-harm in disaster-affected communities.

Design 2 — Action research in one district, co-designing and testing a small-scale ILRM.

Site: One district where the scholar has practitioner access, e.g., a Brahmaputra tributary catchment (Jiadhal, Subansiri) or an eastern UP terai cluster.

RQs and hypotheses: Can a co-designed integrated EWS+NbS package, deployed at a sub-block scale (5–15 villages), produce measurable improvements in lead time, warning equity, and small-event hazard attenuation within 18–24 months? H1: warning reach with comprehension improves by ≥ 30 percentage points; H2: time-to-evacuation falls by $\geq 40\%$; H3: peak depth in the treated micro-catchment falls by 5–15% for design events under modelled scenarios.

Data required: same as Design 1 plus pre/post intervention monitoring, real-time sensor data, structured drill performance data.

Methods: participatory action research (Reason & Bradbury tradition), with a longitudinal mixed-methods evaluation; modelling for NbS hazard attribution; ethnography during one or two flood seasons.

Ethics and practicalities: heightened — the scholar is intervening, not just observing. Strong consent protocols; community-led design; equity safeguards (no benefit capture by elites); transparent communication of uncertainty.

Design 3 — Evaluation of an existing program with a focus on last-mile equity.

Site: A district where a state or NGO program is already operating an EWS or NbS intervention.

RQs and hypotheses: How effective is the existing program in delivering equity-disaggregated risk reduction, and what are the mechanisms of inclusion and exclusion? H1: warning reach is high in aggregate but materially lower for women-headed, SC/ST, tenant, and PwD households; H2: NbS benefits accrue disproportionately to landowning households unless institutional design corrects for it.

Data required: program records, beneficiary lists, household survey, qualitative interviews, GIS overlays of NbS works on cadastral and social maps.

Methods: equity audit using a structured framework (e.g., GenCap / Sphere / Sendai gender markers adapted to flood DRR); qualitative process-tracing; institutional analysis.

Ethics: independence from program implementer; explicit protocols against retaliation; anonymisation.

Recommendation: Design 1 (comparative) is the most defensible for a PhD because it produces generalisable evidence, has clear external validity, and aligns with funder logic. Design 2 (action research) is highest-impact for practice but riskier on dissertation completion if the intervention is delayed or disrupted. Design 3 is feasible if Design 1's data access is constrained. A *hybrid Design 1 + 3*, where the scholar also serves as a thoughtful evaluator of an ongoing program in one of the two comparison districts, often produces the richest dissertation.

6.2 Open data and accessible tools — a minimum viable research design

Hydrometeorological and remote-sensing data (free or low-cost):

- IMD gridded rainfall (0.25° daily, 1901–present); IMD AWS where accessible; CWC daily and 3-hourly discharge bulletins.
- CHIRPS (1981–present, 0.05°, daily, free) for rainfall climatology and gap-filling.
- GPM-IMERG (0.1°, half-hourly) for event-scale precipitation.
- ERA5 reanalysis (ECMWF Copernicus) for atmospheric drivers.
- Sentinel-1 SAR (10 m, 6–12 day) — the workhorse for flood extent mapping in cloudy monsoon conditions; freely processed in Google Earth Engine.
- Sentinel-2 (10–20 m, optical) for land use, vegetation, NbS condition.
- MODIS NRT flood products for large-scale historical events.
- SRTM-DEM (30 m), ALOS-PALSAR DEM (12.5 m), and Bhuvan CartoDEM where licensable, for hydraulic modelling.
- WorldPop and HRSL for population exposure.
- OpenStreetMap for infrastructure.

Open-source models:

- *HEC-RAS* (USACE, free) — 1D/2D hydraulic modelling.
- *HEC-HMS* — rainfall-runoff modelling.
- *SWAT/SWAT+* — catchment-scale hydrology and NbS scenario testing (especially upstream catchment treatment).
- *LISFLOOD-FP* — fast 2D hydrodynamic model, well-suited to large floodplains.
- *CaMa-Flood* — global river hydrodynamics, useful for basin context.
- *QGIS + Google Earth Engine* — analysis environment.
- *PCRaster, gdal, rasterio, xarray* — Python ecosystem for spatial analysis.

Communication technologies for last-mile EWS (low-cost):

- Bulk SMS gateways and IVR (■0.05–0.20 per message at scale).
- WhatsApp Business cascades through ward-level nodes.

- Community FM radio (where licensed); All India Radio Vividh Bharati district programmes; community announcement systems (mosque, temple, gurdwara loudspeakers, with consent).
- Solar-powered sirens at panchayat headquarters (■50,000–2,00,000 per unit, installation included).
- Volunteer cadres (Aapda Mitra under NDMA, women's SHG cadres, Yuvak Mandals, NSS, NCC).
- Manual gauge boards and colour-coded river marks (■2,000–10,000 each, near-zero maintenance).
- Low-cost telemetered water-level sensors (open-hardware ultrasonic units around ■5,000–25,000).

A "minimum viable" research stack for a part-time PhD looks like this: IMD + CHIRPS rainfall × Sentinel-1 SAR flood extent × SRTM DEM × HEC-RAS 2D for the pilot reach × SWAT for the upstream catchment × Google Earth Engine for change detection × QGIS + Python for analysis × structured household survey (KoboToolbox) × NVivo/Taguette for qualitative coding × Excel/R/Python for DiD or synthetic control.

This stack is achievable on a laptop, at a total *software* cost approaching zero, and a *data* cost of zero for most layers. The scholar's effort budget should focus on careful field design, partnerships, and writing.

C. Consolidated Building-Blocks Table

The matrix below presents the full architecture of an Integrated Last-Mile Resilience Model (ILRM) as a single reference table. Rows are building blocks; columns are *Description*, *Why it matters for risk reduction*, *Scalability considerations*, and *Data needed to evaluate*.

Table C.1 — Building blocks of an integrated district-scale EWS + NbS model

Building block	Description	Why it matters for risk reduction	Scalability considerations	Data needed to evaluate
EWS — Risk knowledge layer	Hazard atlases, historical flood maps, community risk maps, vulnerability profiles, exposure layers	Foundational; without shared risk knowledge, warnings are not interpretable and NbS targeting is poor	Standardised national-format atlases (NDMA, NRSC), open formats, periodic update protocol	NRSC Bhuvan flood atlas, IMD climatology, SECC/Census, CEEW CVI, participatory map outputs
EWS — Monitoring & forecasting	Rain gauges, river gauges (manual + telemetered), weather radar, AWS, hydrological + hydrodynamic forecast models	Determines forecast skill and lead time; the scientific backbone of the warning chain	National backbone (CWC, IMD) + state densification + community-operated gauges; open APIs	POD, FAR, CSI, RMSE; gauge density per 1000 km ² ; data uptime
EWS — Warning dissemination	SMS, IVR, WhatsApp cascade, FM radio, sirens, door-to-door volunteers, community boards	Determines warning reach and equity; the most common point of last-mile failure	Multi-channel redundancy; volunteer cadre standardised through NDMA; equity protocols mandated	Channel logs, household survey on reach and comprehension; equity-disaggregated reach
EWS — Response capability	Evacuation plans, shelters, livestock pairing, anticipatory cash, drills, SDRF/NDRF readiness	Converts warnings into protective action; the decisive determinant of avoided losses	Standard SOPs through NDMA; integration with social protection (PMAY, PDS, NREGS); annual drills	Time-to-evacuation, shelter occupancy, anticipatory cash uptake, drill performance
NbS — Upstream catchment treatment	Soil-and-moisture conservation, contour bunding, check dams, agroforestry, watershed development	Reduces and delays peak runoff at small-to-moderate events; livelihoods + groundwater co-benefits	MGNREGS convergence is the single biggest scaling lever; PMKSY-WDC and Atal Bhujal alignment	Land cover change, SWAT outputs, sediment yield, groundwater levels, MGNREGS person-days
NbS — Wetlands & floodplains	Wetland and tank restoration; floodplain reconnection; room-for-the-river style setbacks	Storage and attenuation of flood peaks; biodiversity, water quality, fisheries	Requires spatial planning instruments and tenure clarity; convergence with Wetland Rules 2017, Namami Gange, MISHTI	Wetland area and condition, peak-stage modelling, biodiversity surveys, fisheries data
NbS — Riparian & green buffers	River-edge vegetation belts, urban green-blue infrastructure, sponge-city retrofits	Velocity reduction, bank stability, urban runoff retention; heat island and air-quality co-benefits	Urban: AMRUT 2.0, Smart Cities, municipal master plans; rural: forest department + panchayat	Buffer width and condition, urban impervious surface, runoff retention modelling

Building block	Description	Why it matters for risk reduction	Scalability considerations	Data needed to evaluate
NbS — Coastal NbS	Mangrove and bioshield restoration, dune restoration, salt-marsh and seagrass	Storm-surge attenuation, fisheries, blue carbon	MISHTI, MOEFCC mangrove cell, state forest departments, fishing community stewardship	Mangrove area, density, species composition, surge modelling, fisheries records
Community institutions	Village resilience committees, women's SHGs, Aapda Mitra cadre, Yuvak Mandals	Vehicle for last-mile reach, equity, and stewardship of NbS	Standard NDMA cadre + state Aapda Mitra + integration with NRLM SHG federations	Cadre composition by gender/caste; meeting frequency; activity logs
District governance	DDMA leadership, DDMP integration, line-department coordination, M&E cell	Determines whether the model persists across staff turnover	Mandated DDMP standards; Sendai Framework Priority 2 alignment; state finance commission lines	DDMP audit, budget allocations, coordination meeting records
State governance	SDMA, climate cell, SAPCC integration, recurring budget lines, technical agency	Vertical scaling and policy persistence	SDMA capacity; SAPCC update; SDRMF alignment; central scheme convergence rules	SDMA scorecard, state plan budgets, SAPCC indicators
Indicator & MIS system	Real-time dashboard with hazard, exposure, EWS performance, NbS condition, equity layers	The "instrument panel" that enables evidence-based management and learning	Open-source MIS, interoperable with SDMA, central NDMA Sahayata, IDS-DRR	Dashboard uptime, indicator frequency, audit trails
Equity & gender architecture	Gender markers, equity-disaggregated indicators, women-led committees, PwD inclusion	Without explicit equity design, warnings reach the connected and bypass the most vulnerable	Mandated through GCF/AF/World Bank gender policies; SAPCC gender annex	Equity gap ratios, women's leadership share, PwD-inclusive evacuation drills
Anticipatory finance	Pre-arranged cash transfers triggered by forecast thresholds; livestock insurance; risk pools	Converts forecast skill into pre-event protective action; reduces post-event distress sales	DBT pipes via Aadhaar; integration with PM-KISAN, PMAY, social protection	Trigger fidelity, transfer timing, household uptake, post-event welfare
Capacity development	NIDM/SIRD/ATI curricula; women's SHG training; technical training for DDMA staff	Sustained human capital is the binding constraint to scaling	Standard curricula; certification; recurring budget through NIDM and SDMI/ATI	Training records, competency tests, post-training application metrics
Funding architecture	Blended public + multilateral + philanthropic + CSR + concessional debt	Sustained finance is the difference between pilots and persistence	Convergence rules (e.g., MGNREGS + NAFCC + state plan); blended-finance vehicles	Budget execution, leverage ratio, sustainability index

Building block	Description	Why it matters for risk reduction	Scalability considerations	Data needed to evaluate
Research and learning arm	Embedded MERL with research partner (PhD, university, think-tank)	Generates evidence that wins next round of funding and informs policy	Standing partnership agreements with universities and CSE/ CEEW/IWMI/ATREE	Peer-reviewed outputs, policy uptake, citation in DDMP/SAPCC updates

D. A Tentative Four-Year PhD-Style Work Plan

This plan assumes Design 1 (comparative two-district study with a hybrid evaluation lens) as the primary scaffold, with action-research elements incorporated where the practitioner role enables co-design.

Year 1 — Literature, framing, scoping, and protocol

Months 1–4: Literature and framework consolidation. Systematic literature review using PRISMA-guided protocol on community flood EWS, NbS for floods, integrated DRR-CCA programs, and equity in EWS. Database mining (Scopus, Web of Science, Google Scholar, ScienceDirect, Sendai Framework reports, IPCC AR6, UNDRR Words into Action guidelines, WMO publications, CWC and IMD technical reports, CEEW/IWMI/IIT working papers). Output: a publishable systematic review (target journals: *Climate Risk Management*, *International Journal of Disaster Risk Reduction*, *WIREs Climate Change*).

Months 3–8: Conceptual framework and indicator system finalisation. Operationalise the IPCC risk identity into a measurable framework. Build the indicator dictionary aligned to Sendai Targets, SDGs, GCF result areas. Pre-register the evaluation hypotheses on OSF (Open Science Framework) — a credible discipline that funders and reviewers respect.

Months 6–10: Site selection and partnership formation. Negotiate MoUs with two SDMA (e.g., Assam SDMA + Bihar SDMA), one technical partner (IIT/NIH/IIRS), and one or two CSO partners with field presence. Secure ethics clearance from the host university IRB. Design and pilot survey instruments. Build the data acquisition pipeline (Bhuvan, Sentinel, GEE workspace).

Months 9–12: Reconnaissance fieldwork and proposal defence. Two short field visits to each district (4–6 weeks total). Participatory hazard mapping in 4–6 villages. Stakeholder mapping. Defend the PhD research proposal. *Year 1 deliverables:* systematic review paper (submitted), framework paper (drafted), proposal defended, MoUs in place, OSF pre-registration.

Year 2 — Fieldwork, data collection, baseline establishment, pilot design

Months 13–16: Baseline survey and institutional analysis. Household surveys ($n \approx 600$ –1000 across treatment and comparison clusters in both districts) using KoboToolbox. Key informant interviews with DDMA, SDMA, panchayat, line departments, CSOs ($n \approx 40$ –60). Build the *Equity-Disaggregated Vulnerability Layer* (a methodological contribution).

Months 14–18: Hydrological and remote-sensing baseline. Build a 10–15 year flood-frequency map for each district from Sentinel-1 SAR archive (GEE). Set up HEC-RAS 2D models for selected reaches. Calibrate SWAT for the upstream catchment of one district.

Months 17–22: Monsoon-season monitoring and after-action research. First full monsoon under observation. Document warning chains, evacuations, losses for any events. Conduct after-action reviews with DDMA, panchayats, communities. This is when the *warning-to-action gap* becomes empirically visible.

Months 20–24: Pilot intervention design (if Design 1 is augmented with action-research). Co-design with one cluster of 4–6 villages (in the partner district): an integrated package — community gauge readers, multi-channel warning cascade with women's SHG nodes, NbS micro-interventions (riparian buffer planting, tank-cascade restoration through MGNREGS), and equity-disaggregated indicator dashboard. *Year 2 deliverables:* baseline data paper (drafted), institutional analysis paper (drafted), monsoon-season case study, intervention design document.

Year 3 — Implementation, monitoring, mid-term analysis

Months 25–28: Intervention rollout in pilot cluster; intensified monitoring. Deploy the package. Establish telemetered gauges. Train and certify the cadre. Pre-register the anticipatory action protocol

with the DDMA.

Months 27–34: Second-monsoon evaluation. Track warning lead time, reach, equity, and (where events occur) avoided losses. Hydraulic modelling of with-and-without scenarios. Sentinel-1 flood mapping pre/during/post. Quasi-experimental analysis (DiD or synthetic control) of event outcomes.

Months 30–36: Mid-term analysis and writing. First full empirical paper: comparative effectiveness of EWS + NbS in two districts (the *RQ1* paper). Second paper: equity audit and warning-to-action mechanisms (*RQ2*). Mid-term presentation at international conferences (UNDRR Global Platform side events; AGU; AOGS; IUCN; APAN).

Year 3 deliverables: two empirical papers submitted; mid-term review with university; policy brief shared with both SDMAs; project report to funders.

Year 4 — Synthesis, scalability analysis, dissemination and policy engagement

Months 37–40: Synthesis chapter and scalability paper. Write the synthesis: the *RQ3* paper on scalability architecture. Build a costed replication template usable by SDMAs and funders. Prepare a *funder-ready blueprint document* that translates the PhD into IQ1 — the practical "what should be done in vulnerable Indian districts".

Months 39–44: Dissertation finalisation. Integrate empirical chapters, methodology, framework, and synthesis into the PhD thesis. Internal review with supervisors. External review.

Months 42–48: Defence, dissemination, policy engagement. Defend. Publish remaining papers. Convene a national workshop (with NIDM/CEEW/NDMA). Engage with state SDMAs on adoption. Submit a concept note to a major funder (GCF, AF, or World Bank) drawing directly on the dissertation. Optionally, formalise a 3–5 year follow-on research-implementation program.

Year 4 deliverables: dissertation defended; 4–5 peer-reviewed papers published or accepted; one funder concept note submitted; one state-level policy adoption (e.g., DDMP guideline update); one international conference presentation; a dissemination plan including policy briefs in English and one regional language.

Honest caveats on the work plan

- Part-time scholarship: stretch each phase by 25–40% if the scholar's practitioner workload is substantial. A 5-year horizon is realistic.
- Disaster context: at least one extreme event during the four years is likely to disrupt the field plan; build in a six-month buffer.
- Data risk: CWC discharge data access remains uneven across basins; secure formal data-sharing MoUs early.
- Ethical risk: working with vulnerable communities requires standing relationships, not transactional access; the practitioner role is an asset here.
- Political risk: state political turnover can reset SDMA priorities; build relationships with permanent technical staff (additional secretaries, joint commissioners) in addition to political leadership.

E. Open Research Gaps and Unresolved Debates Suitable for PhD Contribution

A defensible PhD makes a *contribution to knowledge*. The integrated EWS+NbS field has at least five live debates where a rigorous India-grounded PhD can move the literature.

1. **Causal attribution of NbS hydrological effects under non-stationary climate.** Most NbS evidence is from temperate basins or Bangladesh coasts; tropical-monsoonal basin attribution under climate change is sparse and contested. A PhD that combines paired-catchment monitoring with modelling and Bayesian uncertainty quantification offers a real contribution.
2. **Inclusion-weighted indicators of EWS reach.** Current frameworks measure *coverage*, not *equity*. A methodological contribution that operationalises a *warning-equity index* and demonstrates its application in two Indian districts would be widely cited.
3. **The warning-to-action behavioural gap.** Mixed-methods evidence on what specifically converts a received warning into household action — particularly for women, Dalit, Adivasi, and tenant households — is thin in India. Ethnography combined with survey-based behavioural analysis is publishable in *Disasters*, *Global Environmental Change*, *World Development*.
4. **Governance and finance models for sustained district-level EWS+NbS.** The political economy of *why* states under-invest in DRR despite high benefit-cost ratios remains under-theorised in India. A comparative governance analysis using the Ostrom IAD framework or the Sendai governance lens would be a valuable contribution.
5. **Climate-change-adjusted cost–benefit frameworks for NbS that internalise co-benefits.** Standard CBA undervalues NbS by ignoring biodiversity, carbon, and livelihood co-benefits, and by using stationary climate baselines. A methodological contribution updating CBA for non-stationary climate and multifunctional benefits, applied to one Indian district, is timely and policy-relevant.

A PhD that addresses 2–3 of these gaps — even partially — through the comparative two-district design will be a strong contribution to both literature and practice. The argument of this brief is that this is achievable, ethical, and impactful, and that it can be completed in four years (or five for part-time) with the partnerships and tools described above.

F. Sources and Reference Frameworks (indicative, not exhaustive)

The brief draws on the following bodies of evidence and frameworks. Specific citations should be added during the literature review (Year 1 of the work plan); the list below names the categories and key institutional sources to consult, rather than fabricating page-precise citations.

- **Multilateral frameworks:** UNDRR Sendai Framework for Disaster Risk Reduction 2015–2030 and Midterm Review (2023); UN Secretary-General Early Warnings for All Action Plan (2022, with 2023 implementation roadmap); WMO Multi-Hazard Early Warning System guidelines; IPCC AR6 WGII (2022) and Synthesis Report (2023); IUCN Global Standard for NbS (2020); UNEA-5 NbS resolution (2022); CBD Kunming-Montreal GBF (2022).
- **Indian institutional sources:** NDMA hazard atlases and DDMP guidelines; CWC flood forecasting bulletins and basin reports; IMD rainfall and AWS data; NRSC Bhuvan flood atlas; CEEW *Climate Vulnerability Index of Indian States and Districts* (2021); DST/SDC *Climate Vulnerability Assessment* reports (2019, 2021); MoEFCC mangrove and wetland reports; Ministry of Jal Shakti reports; Ministry of Rural Development MGNREGS guidelines; State SDMA reports and DDMPs.
- **Academic literature:** peer-reviewed papers in *Natural Hazards and Earth System Sciences*, *Climate Risk Management*, *International Journal of Disaster Risk Reduction*, *Disasters*, *Global Environmental Change*, *Hydrology and Earth System Sciences*, *Nature-Based Solutions*, *Environmental Research Letters*, *World Development*, *Regional Environmental Change*.
- **Practitioner evidence:** independent evaluations of CPP Bangladesh, Practical Action Nepal/Peru CBEWS, ADPC Mekong, Red Cross/IFRC anticipatory action, World Bank ICR/IEG reviews of NCRMP and NHP, GCF and AF project performance reports.
- **Indian programs and case bodies:** Aaranyak (Assam EWS), SEEDS, Practical Action South Asia, Goonj, Welthungerhilfe, ATREE, ICRISAT, IWMI, CSE (Centre for Science and Environment), CEEW, NIDM and ATIs, IITs (Roorkee, Guwahati, Madras, Bombay, Gandhinagar), IISc, IIRS, NIH-Roorkee, NIO/NCCR for coastal.

A first-pass systematic review (PRISMA-guided) over 6–8 weeks of dedicated work in Year 1 will produce a structured, citable bibliography of approximately 250–400 references.

End of brief. This document is intended as a working scaffold for a PhD inquiry plus a funder-attractive practical model. It should be revised iteratively as the literature review and fieldwork progress, and as funders' priorities and policy windows evolve.